

Erxiao Wang

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RESEARCH INTERESTS

Neural representation of language; incremental sentence comprehension; language-speech-motor interfaces; bilingual neural plasticity.

EDUCATION

Columbia University, Teachers College, M.S. in Biobehavioral Neuroscience New York, NY, 05/2024

- *Thesis: Multimodal Speech Processing in the Bilingual Brain: A Systematic Review of the Literature*
- *GPA: 3.91/4.0*

Wenzhou-Kean University, B.A. in English (focus on education and linguistics) Wenzhou, China, 06/2021

- *Coursework: Introduction to Linguistics, Historical Linguistics, Sociolinguistics, Educational Linguistics*
- *GPA: 3.86/4.0 (Summa Cum Laude)*

PUBLICATIONS

1. **Wang, E.**, & Shain, C. (2025). Repetition Increases Signal in Self-Paced Reading Experiments. (paper in prep) [[link to OSF](#)]
2. Zhang, Y., **Wang, E.**, & Shain, C. (2025). Graded strength of comparative illusions is explained by Bayesian inference. (paper submitted) [[preprint](#)]

RESEARCH EXPERIENCE

Laboratory for Computation and Language in the Mind and Brain

Stanford University, Department of Linguistics · Stanford, CA

Lab Manager / Research Coordinator, full-time

09/2024 to present

P.I.: Dr. Cory Shain

- Analyze large-scale self-paced reading data with continuous-time deconvolutional regression (CDR) and cross-validated decoding to examine stimulus-driven sentence-processing structure and adaptation under verbatim repetition.
- Collect and analyze intensive individual-level fMRI data for language and speech-motor mapping in the INT project and related NPTL collaboration. Use RSA to examine response-pattern geometry and gPPI to characterize task-dependent functional coupling.
- Investigate comparative illusions with Dr. Yuhua Zhang using noisy-channel Bayesian inference over plausible intended meanings and behavioral acceptability judgments.
- Analyze intracortical microelectrode array recordings during naturalistic listening and reading using generalized additive models (GAMs), feature decoding, and principal component analysis (PCA) to characterize linguistic encoding and population dynamics in language- and speech-sensitive regions.
- Assist with IRB submissions, participant coordination, and general lab operations.

Hartley Lab, Psychology Department, New York University
Graduate Research Associate, part-time

New York, NY
10/2023 to 08/2024

P.I.: Dr. Catherine Hartley

- Engineered a robust Python-Flask server with an SQL database to automate longitudinal data collection and SMS-based participant communication, enabling randomized multi-modal data scheduling over three-month study periods.
- The system has been operating sustainably and has supported data from 100+ participants, significantly reducing manual coordination efforts. This system facilitated research on how geographical novelty interacts with episodic memory across development.
- Programmed experimental tasks using JavaScript to study memory integrity and valence-biased reinforcement learning across development.

Social Cognitive and Neural Science Lab, Psychology Department, Columbia University
Research Assistant, on-campus

New York, NY
09/2023 to 08/2024

P.I.: Dr. Jon Freeman

- Assisted in coding fMRI-compatible experiments and collecting data for studies on the neural basis of racial identity and facial perception.

Language and Cognition Neuroscience Lab, Teachers College, Columbia University
Research Assistant, on-campus

New York, NY
09/2022 to 05/2024

P.I.: Dr. Peter Gordon

- Developed multimodal EEG and eye-tracking paradigms to study the role of language and symbol mapping in numerical cognition among monolingual Korean and bilingual Mandarin-English participants.
- Collected and analyzed data for developmental numerical cognition with Mandarin-speaking 2-4-year-old children.

POSTERS & PRESENTATIONS

1. [Poster] **Wang, E.**, Zhang, Y., Madugula, S., Kunz, E. M., Vo, C., Avansino, D. T., Wang, R., Marin Vargas, A., Ergin, I., Rojas, R. R., Hochberg, L., Gwilliams, L., Henderson, J. M., Willett, F. R., & Shain, C. (November 14–18, 2026). Distinct temporal profiles in middle frontal gyrus (MFG) and inferior frontal gyrus (IFG) during naturalistic comprehension. Neuroscience 2026, Society for Neuroscience Annual Meeting, Washington, D.C. Abstract 6815.
2. [Poster] Zhang, Y., **Wang, E.**, Madugula, S., Kunz, E. M., Vo, C., Avansino, D. T., Wang, R., Marin Vargas, A., Ergin, I., Rojas, R. R., Gwilliams, L., Hochberg, L. R., Henderson, J. M., Willett, F. R., & Shain, C. (November 14–18, 2026). Intracortical evidence for functional dissociation from word encoding to meaning integration within the language network. Neuroscience 2026, Society for Neuroscience Annual Meeting, Washington, D.C. Abstract 6880.
3. [Talk] **Wang, E.**, Ivshina, E. S., Zhang, Y., & Shain, C. (March 26–28, 2026). Sentence processing is surprisingly robust to verbatim repetition. The 39th Annual Conference on Human Sentence Processing, Massachusetts Institute of Technology, Cambridge, MA.
4. [Talk] Zhang, Y., **Wang, E.**, & Shain, C. (November 14–16, 2025). Graded strength of comparative illusions is explained by Bayesian inference. The 8th California Meeting on Psycholinguistics, University of California, Davis, CA.
5. [Talk] **Wang, E.**, & Shain, C. (November 2025). Repetition Increases Signal in Self-Paced Reading Experiments. *The 8th California Meeting on Psycholinguistics (CAMP8)*, Davis, CA.

6. [Poster] Benear, S.L., **Wang, E.**, Hartley, C.A. (August 2025). The effects of real-world novelty exposure on episodic memory specificity across development. *Cognitive Science Society Conference (CogSci)*, San Francisco, CA. [[link to poster](#)]
7. [Poster] Zhang, Y., **Wang, E.**, & Shain, C. (March 2025). Distance to plausible alternatives predicts acceptability ratings in comparative illusion. *The 38th Annual Conference on Human Sentence Processing*, College Park, MD. [[link to poster](#)]
8. [Poster] Oh, E., **Wang, E.**, Xiong, W., Cha, M. & Gordon, P. (May 2023). Two Number Systems with Bilinguals and Monolinguals: An ERP Study of Code-Switching. *Association for Psychological Science (APS) Annual Convention*, Washington DC.

TEACHING EXPERIENCES

Developmental Cognitive Neuroscience, Course Assistant, Teachers College, Columbia University (2024 Spring)

Second Language Acquisition Seminar, Teaching Assistant, Wenzhou-Kean University (2021 Spring)

HONORS & AWARDS

Biobehavioral Sciences Department Scholarship, 2024

Provost Travel Grant, 2023

Student-Faculty Research Partnership Grant, 2020

Dean's List Scholarship, 2018–2019

SERVICES & LEADERSHIP

Organizer, *Columbia University Neuroscience Outreach (CUNO)* – Partnered with local education organization to provide accessible hands-on science activities for local neighborhoods, especially for underrepresented families and children (2023).

EEG Training Program Mentor – Supervised and trained students from the Language and Cognition Neuroscience Lab in EEG experimental setup and analysis (2023–2024).

Academic Peer Tutor – Coached over 150 individual sessions for students under academic alert and facilitated large-group workshops on academic presentation skills (2018–2021).

Founder and President, *French Club* – Established and led a student group promoting French language and culture through learning and cultural events (2018–2019).

SKILLS

Computational: Python, MATLAB, R, JS, Git, UNIX command-line tools, Slurm jobs.

Statistics and modeling: Linear mixed-effects models, Bayesian regression, continuous-time deconvolutional regression (CDR), generalized additive models (GAMs), cross-validated decoding, principal component analysis (PCA), machine learning (PyTorch, scikit-learn), Hugging Face Transformers.

Research methods: fMRI and EEG data collection and analysis (fMRIPrep, Nilearn); intracortical recording analysis; representational similarity analysis (RSA); generalized psychophysiological interaction analysis (gPPI); reusable analysis pipelines; online behavioral experiments (jsPsych, Prolific).

Language: native in Mandarin; fluent in English; elementary proficiency in Spanish.